

Significance methodology and source-independence review

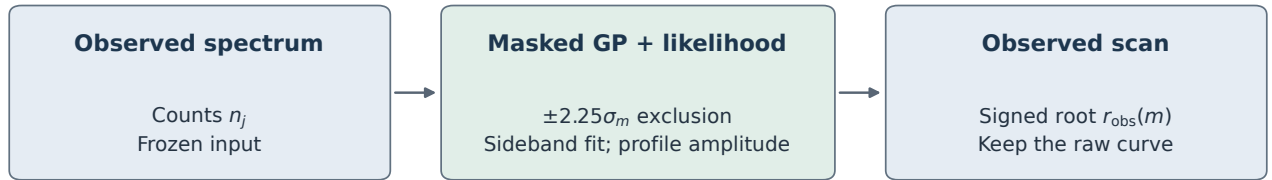
HPS Gaussian-Process Resonance Search
Consolidated Analysis Report v5.8.5

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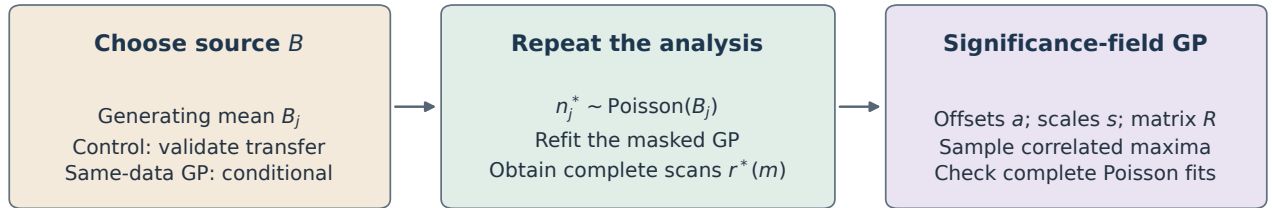
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This report brings together v5.8.0–v5.8.5 and revisits the distinction between the observed likelihood, its local null distribution, and the look-elsewhere effect. The moving blind half-width remains $2.25\sigma_m$. No production significance or limit is replaced by the conditional source studies.

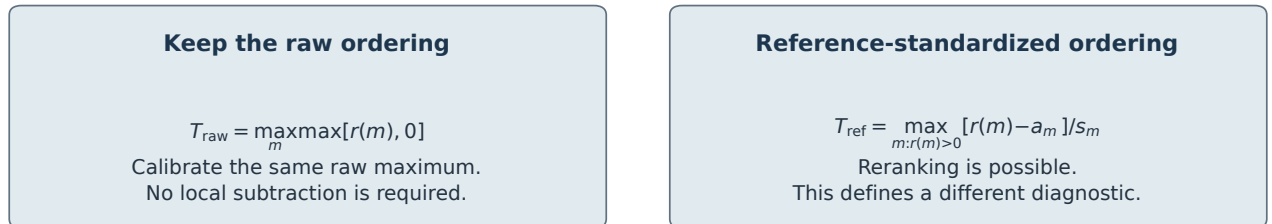
THE OBSERVED ANALYSIS



THE NULL EXPERIMENT AND ITS EMULATOR



SAME DECLARED TEST FOR OBSERVED AND NULL SCANS



A shared source is not an independent background measurement.

Figure 1: The two Gaussian processes and the choice of statistical ordering. The background GP is part of the fitted analysis. The significance-field GP approximates the distribution of that analysis under a specified source. It need not be independent of the background model. Independent evidence is needed to qualify the source and to validate the approximation. Raw and reference-standardized maxima are different tests, even with identical observed events and likelihood fits.

1 What this review settles

The background GP and the significance-field GP can be used in the same analysis without circular reasoning merely because both are Gaussian processes. They have different jobs. The first estimates the background inside the moving mask; the second reproduces the correlated fluctuations of the resulting fitted scan. Their agreement with complete Poisson fits tests the latter approximation. It does not independently establish that the spectrum used to generate those fits is a physical background-only source.

The concern about using a source fitted through the observed events is therefore well founded. Such a source can be used for a conditional diagnostic, or within a carefully validated procedure that estimates background parameters from the data. Here, however, the full source is frozen in the toys, and the earlier injection experiment shows substantial absorption when it is rebuilt through an injected peak. A smoother stress response is useful evidence about a particular reference, but is insufficient grounds for adopting that reference as the production null.

1.1 The quoted 2016 change is more than a trials correction

The raw observed maximum is still $r = 3.4525$ at 90.5 MeV. The reported $Z = 2.886$ at 91.5 MeV is the maximum after a mass-dependent transformation, $z_{\text{ref}} = (r - a)/s$. Calling these merely two summaries obscured a consequential change of statistical ordering. The reference score is a diagnostic calibration relative to the chosen source; it was not needed just to count search opportunities, and it is not adopted as the primary local result here.

A new audit retains the raw ordering and compares it directly with the saved null maxima. Conditional on the same nominal GP source, the 2016 raw-maximum tail is 0.0678 in the Gaussian ensemble, with 18 exceedances in 256 complete Poisson scans. The reference-standardized maximum instead gives 0.1049, with 36/256 direct exceedances. Both are source-conditional probabilities for different tests. The smaller one cannot be selected after inspecting the comparison and then described as an independently specified discovery result.

1.2 The look-elsewhere correction is not an arbitrary conservatism choice

A low correlation at one chosen spacing does not divide a continuous search into independent signal regions. A region still contains a maximization, and correlations can return or become negative at greater separations. A Šidák calculation becomes a meaningful approximation only after the local and regional test events have been specified and the dependence among those events checked.

The paper's improvement concerns an upcrossing upper bound for the same significance field[1]. It does not promise a smaller penalty than counting resolution intervals. The distinction is material for HPS: the fitted response includes sideband subtraction and background retraining, which can produce a more rapidly varying score than a signal template on a known background. The new comparisons below use identical thresholds and covariance matrices wherever possible.

1.3 What remains fixed and what was newly tested

All new calculations retain the $\pm 2.25\sigma_m$ moving mask. The earlier width scans are summarized as sensitivity tests; they provide no reason to choose a new width now. This review adds a raw-versus-reference maximum audit, a covariance-based test of the proposed ten-percent correlation rule, and a bounded 2021 high- p_{sum} source-transfer diagnostic. It does not repeat the previous full fit scans or claim a new rare-tail calibration.

2 What the 5.8 series has established

The studies answer different questions in a sequence: the old stress reference was diagnosed, a coherent GP source was constructed, the fitted-score covariance was tested under that source, and several combined-search alternatives were compared. The improved stress response is useful evidence about the construction. It does not supply a second, independent measurement of the background.

Study	Main result	Interpretation retained here
5.8.0	Full-2016 raw maximum changes from 3.425 at 90 MeV on the integer grid to 3.453 at 90.5 MeV on the half-MeV grid. Stress RMS stays 4.97.	Finer sampling resolves extrema; it does not remove background mismatch. Shorter analysis length scales do not restore closure everywhere.
5.8.1	Full-2016 deterministic RMS falls from 4.967 for the archived stress source to 0.609 for the nominal GP source; half the source length scale gives 1.700.	A better matched source changes the diagnosis. Source rebuilding can also absorb injected signal; small RMS alone does not qualify the null.
5.8.2	Half-MeV response fields describe moderate-tail complete Poisson checks. The 2016 reference-local peak is 2.886 at 91.5 MeV, with global $p = 0.10488$.	Both local reference mapping and the mass-search correction are conditional on the fixed source.
5.8.3	Saved fields explain why full-response correlations give a larger penalty than counting independent FWHM intervals.	An interpretation audit; no new likelihood fits or random experiments.
5.8.4	Integer-grid Fisher and signed Stouffer peaks stay at 92 MeV over the four tested widths. The shared-coupling peak stays near 66 MeV.	Different alternatives and weights select different features. These are repeated analyses of the same spectra.
5.8.5	The free-amplitude likelihood peaks at 91–92 MeV. Baseline conditional local/global $Z = 3.583/2.060$; mass coherence is unexceptional.	A common mass with free nonnegative yields is a distinct hypothesis. Its square-root likelihood ratio is not itself a Gaussian significance.

Table 1: Verified study ledger. The new synthesis retains $\pm 2.25\sigma_m$ as the nominal blind half-width. Numeric inputs, source hashes and important original figure paths are listed in `results/study_ledger.json`.

2.1 The earlier note already separates raw and reference-conditioned tests

Section 4.20 of v5.0.5, on pages 58–61, first defines the signed likelihood root and its conventional asymptotic display. Equations (88)–(93) then define the deterministic offset, response covariance, marginal scale and simulated field. Equations (94)–(96) explicitly introduce a *different*, reference-standardized ordering, with a positive-raw-fit gate. The text warns that this extension does not automatically acquire the discovery interpretation of the Ananiev–Read construction. Section 6.11, beginning on page 114, reports the resulting conditional stress-background curves separately from the asymptotic and Šidák references.

This distinction was essential in the 76 MeV example: the full-2016 raw fit is negative, $r = -2.460$, and the shared-coupling raw fit is only $r = 0.166$. Its roughly nine-unit stress-centered coordinate came from a strongly negative reference response. It was not a nine-standard-deviation observed resonance. The later nominal-source studies removed much of that particular mismatch but retained the same conditional interpretation.

The quoted v5.8.3 paragraph therefore describes an actual reference centering and scaling in a diagnostic probability calculation. It is not merely the look-elsewhere correction, and it should not be read as authorization to replace the released local inference. The observed data, likelihood and fitted raw

roots were unchanged. In this report the raw likelihood result remains visible; the reference-conditioned result is identified as a model-dependent diagnostic.

Mass [MeV]	Raw r	Offset a	Scale s	$(r - a)/s$	Full-range p
90.0	3.42475	0.79076	0.98224	2.68160	0.17712
90.5	3.45250	0.71285	0.98203	2.78978	0.13522
91.0	3.40112	0.63026	0.98157	2.82288	0.12445
91.5	3.33316	0.49674	0.98284	2.88595	0.10488
92.0	3.11542	0.36419	0.98395	2.79610	0.13291

Table 2: The exact v5.8.2 full-2016 mapping behind the quoted paragraph. The final column compares each reference score with the maximum anywhere on the 39–180 MeV half-MeV grid. It is not a local probability.

At 90.5 MeV, $(3.45250 - 0.71285)/0.98203 = 2.78978$. At 91.5 MeV, the smaller offset gives $(3.33316 - 0.49674)/0.98284 = 2.88595$. Thus centering changes which mass ranks highest before any global comparison is applied. The subsequent search correction maps the reference-local 2.88595 to global 1.25423. These are two separate operations; calling their combined effect a trials penalty obscures the source dependence.

Passing prior statistical tests does not require applying this centering to the production result. Nor does it prove the raw root is standard normal under every new generator. The prior empirical two-truth calibration tested its calibrated envelope, whereas the nominal-GP source in v5.8.1 still gave 101/512 raw nominal 5% rejections at 90 MeV and 97/512 at 117 MeV. These tests have different probability constructions and assumptions. Good centered moments in v5.8.2 check fluctuations around its specified source; they do not settle which source or local calibration should define a physical discovery test.

3 Reading Figures 4 and 5 of v5.8.3

3.1 Figure 4: correlation is a property of repeated fitted scores

At a separation of, for example, 5 MeV, each pair of tested masses contributes a correlation between the two fit results produced by repeated fluctuations of the same spectrum. The solid line averages those correlations over all pairs with that separation. The blue band is the 10th–90th percentile spread across mass locations: it is not a confidence band, a range of candidate widths, or a fluctuation in the observed data.

The dashed curve asks a simpler question: how much do two positive Gaussian signal templates overlap when the background is known and the noise is locally white? With equal resolution σ_m , that correlation is $\exp[-\Delta m^2/(4\sigma_m^2)]$. One FWHM of separation still leaves correlation 1/4. Detector resolution therefore does not create sharply bounded independent signal regions.

The real score compares the signal region with a background inferred from surrounding counts. A fluctuation that helps an excess at one mass can raise the inferred background at another and reduce its fitted excess. Such opposite responses produce negative correlations. Moving masks, retraining, profiling, varying counts, resolution and kernel settings all enter the saved response. The negative lobes do not show a negative event rate or prove that a signal has been removed. They also do not isolate the moving mask as the sole cause: that would require a controlled comparison in which only mask movement is changed.

The saved fields have median adjacent half-MeV correlations of 0.9705, 0.9897 and 0.9880 for 2015, 2016 and 2021, yet reach negative correlations near -0.85 to -0.87 farther apart. High correlation between neighboring tests is therefore fully compatible with appreciable opportunities for a maximum across the complete domain. The maximum calculation uses the full covariance matrix, including its

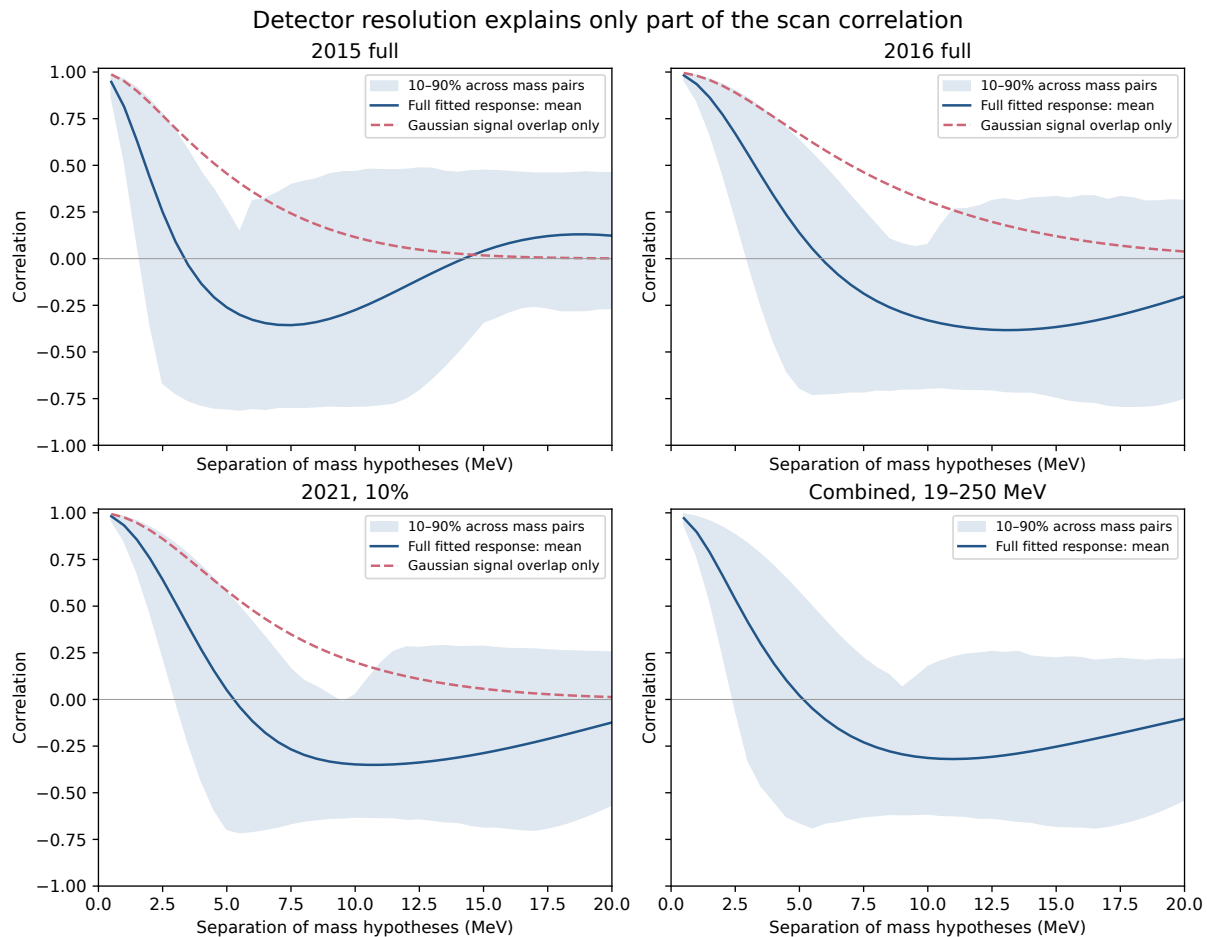


Figure 2: Reproduction of v5.8.3 Figure 4. The fitted-response correlation is compared with Gaussian-template overlap. Shading describes variation across mass pairs, not uncertainty in a measured correlation. These are the saved nominal-source fields, with no new fits.

nonlocal structure; it does not replace it with this averaged lag plot.

3.2 Figure 5: compare the same threshold under different scan models

The horizontal axis is a hypothetical *reference-local threshold*, not mass. At any fixed value, the vertical axis asks how often a background search exceeds that threshold somewhere. The thin black curve gives one specified mass. Gray treats every half-MeV grid point as independent. Orange uses a fixed FWHM-spacing count in the Šidák formula. Green uses the high-threshold approximation for a smooth Gaussian-template scan. Blue is the saved correlated fitted-response calculation; its narrow band covers finite Gaussian sampling only. The dot marks the observed reference-local peak.

For the 2016 threshold $u = 2.88595$, one mass has $p = 0.001951$. The FWHM-count approximation gives 0.02953, the smooth-template approximation gives 0.06521, the fitted field gives 0.10488, and 283 independent grid tests give 0.42461. Thus correlation greatly reduces the penalty relative to independent grid points. The larger probability than the FWHM heuristic does not demonstrate that the method is conservative relative to the correct physical search: the FWHM heuristic is not that calibration. Even the ideal continuous smooth field has more possible high maxima than a fixed width count represents.

The distinction also prevents an incorrect interpretation of “reach.” These are background-only discovery tails. They do not multiply, weaken or otherwise alter the reported fixed-mass 90% CL_s

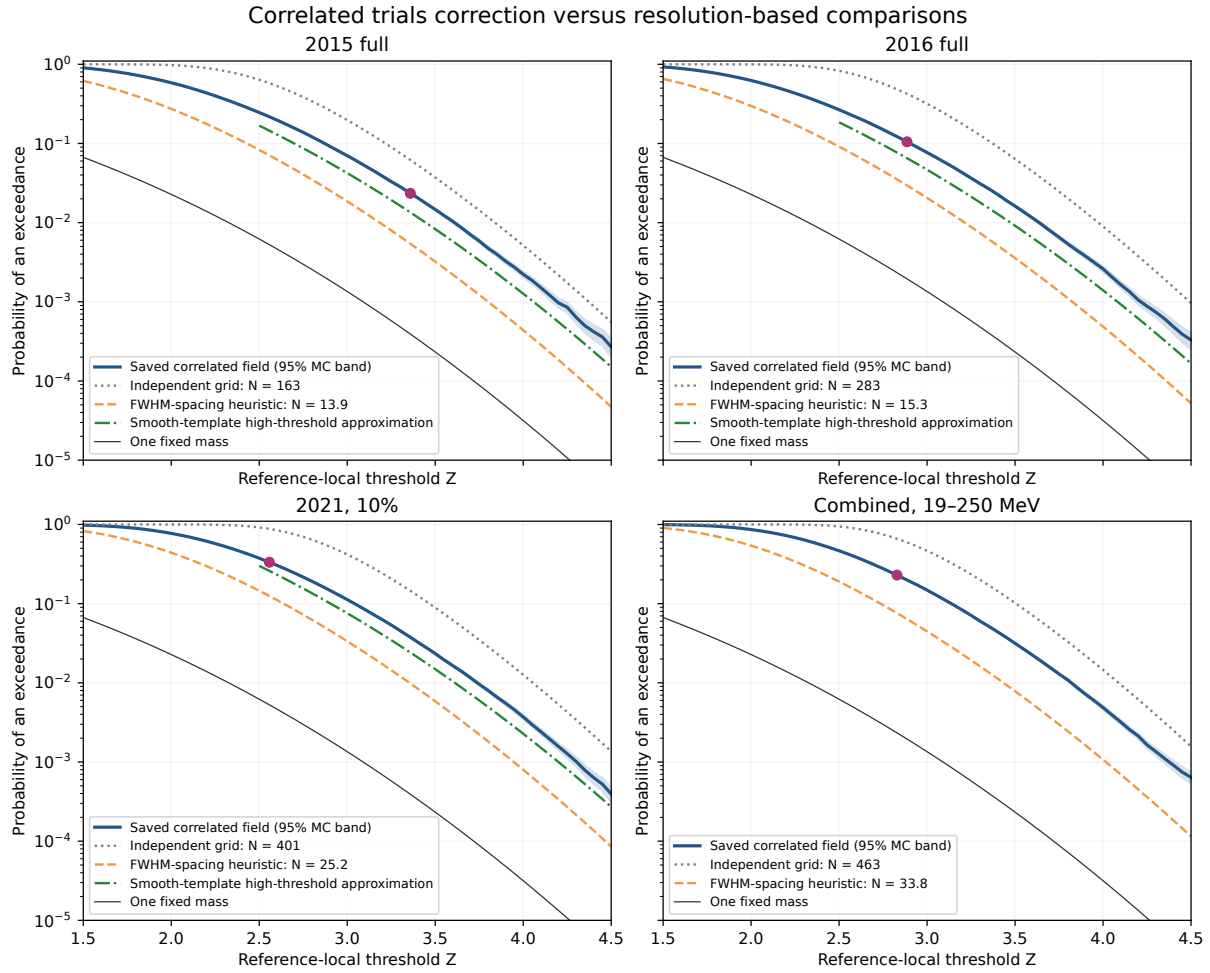


Figure 3: Reproduction of v5.8.3 Figure 5. Every curve in a panel is evaluated at the same reference-local threshold. The gray and orange curves are alternative counting approximations, while green describes an ideal smooth-template field. Blue uses the actual fitted-score covariance under the fixed nominal source. No new observed information enters this comparison.

upper endpoints. A discovery-reach forecast would additionally need signal-plus-background ensembles and a declared global discovery criterion.

4 What survives the combination and width studies

Hypothesis/statistic	Peak [MeV]	Local Z	Global p	Global Z	Direct k
Shared coupling	66	2.829	0.20849	0.812	41
Fisher at common mass	92	3.723	0.01131	2.280	4
Signed Stouffer	92	3.934	0.00494	2.580	1
Free nonnegative yields	92	3.583	0.01968	2.060	7

Table 3: Baseline $\pm 2.25\sigma_m$ results on the 19–250 MeV integer grid, with each dataset used only in its declared range. All significance columns are conditional Gaussian-response results. Counts refer to the same 256 complete Poisson scans; the rows are not independent replications.

At 92 MeV the shared-coupling approximation gives dataset score weights about 0.041, 0.419 and 0.907, whereas signed Stouffer gives each dataset $1/\sqrt{3}$. The independent combinations therefore emphasize the 2015 upward response more strongly. This is a change in the alternative hypothesis or weighting, not a removal of the mass-search correction. The exact independent-yield statistic is $q_R = \sum_d [\max(r_d, 0)]^2$; its local distribution includes the boundary and cannot be replaced by $\sqrt{q_R}$ as a Gaussian significance.

In v5.8.5, seven direct global exceedances give a 95% probability interval [0.0111, 0.0555]. Its baseline local tail has zero direct exceedances in 256 scans, so the reported local 3.583 remains an approximation rather than a directly measured rare tail. The exploratory common-region coherence statistic gives 195/256 scans at least as tightly clustered as the observations, with add-one $p = 0.7626$. Its stable combined maximum provides no separate evidence for unusually aligned individual peaks.

Widths 2.4 , 2.5 and $2.6\sigma_m$ changed the extractor's held-out bins *and* fitted likelihood window, while leaving the generating source fixed. The 2016 source-offset RMS remained about 0.61–0.62. These comparisons therefore did not test whether a wider source-building exclusion removes contamination. One 2016 $2.6\sigma_m$ peak-tail check was flagged: Gaussian $p = 0.09547$ lies just below the direct 95% interval [0.0971, 0.1850]. All configurations remain diagnostics. There is no basis here for changing the nominal $\pm 2.25\sigma_m$ policy, and no old study demonstrated a $3\sigma_m$ source-construction cure.

The apparent numerical shifts between the half-MeV v5.8.2–3 results and the integer-grid v5.8.4–5 results include changed mass sampling and finite Gaussian sampling. They do not represent additional observed events. None of the displayed global probabilities includes choosing the most favorable method, width, dataset subset or mass domain after inspection.

5 What the deterministic offset actually measures

The colored curve in v5.8.2 Figure 3 is $a(m) = r(B; m)$: the signed signal-fit root obtained when the input histogram is replaced by the smooth expected counts B , with no Poisson draw. It is not the observed excess, an estimated number of signal events, or a fraction of signal removed. There are two distinct fits. First, the *generating source* B is a GP fit through every observed bin, with its kernel anchored at 76 MeV. Second, at each tested mass, the ordinary extraction excludes $\pm 2.25\sigma_m$, predicts inside that window from the sidebands, and profiles the signal and background nuisance parameters. The source fit has no moving exclusion window. The extraction has mass-dependent reviewed kernels and count-dependent conditioning. These two operations need not reproduce one another exactly: a smooth B can produce a nonzero signed signal estimate when it is rescanned.

A positive a means that this particular noiseless reference looks signal-like to the extraction at that mass; a negative a means a deficit relative to the extraction's interpolated background. Curvature, interpolation, the different source/extraction kernels and nuisance treatment can contribute. This diagnoses the response to the chosen B ; it does not identify which component of the measured spectrum is continuum or signal. The unchanged observed root remains available separately.

The other two curves in the old Figure 3 are already transformed quantities. Its black curve is $\text{mean}[(r_{\text{toy}} - a)/s]$; its gray dashed curve is $\text{SD}[(r_{\text{toy}} - a)/s]$, where s is the linear-response fluctuation width. They should lie near zero and one if the conditional approximation works. They are neither raw toy roots nor additional offsets to subtract. In particular, a black curve near zero does not imply $a = 0$. The nonlinearity of the scan also means that $r(B)$ need not equal the exact toy mean, although their agreement can be checked.

For example, at 90.5 MeV the stored values are $a = 0.71285$ and $s = 0.98203$. The 256 raw toy roots have mean 0.65653 and SD 1.04126. Their centered mean is about -0.057 , explaining how the old figure can show a sizeable colored offset together with a black curve near zero. At 76 MeV the nominal-GP offset is -0.40839 ; the much larger archived-stress response belongs to a different reference. Neither value alone proves that a particle signal was fitted away.

2016: unpacking v5.8.2 Figure 3 — one frozen GP source, 256 scans

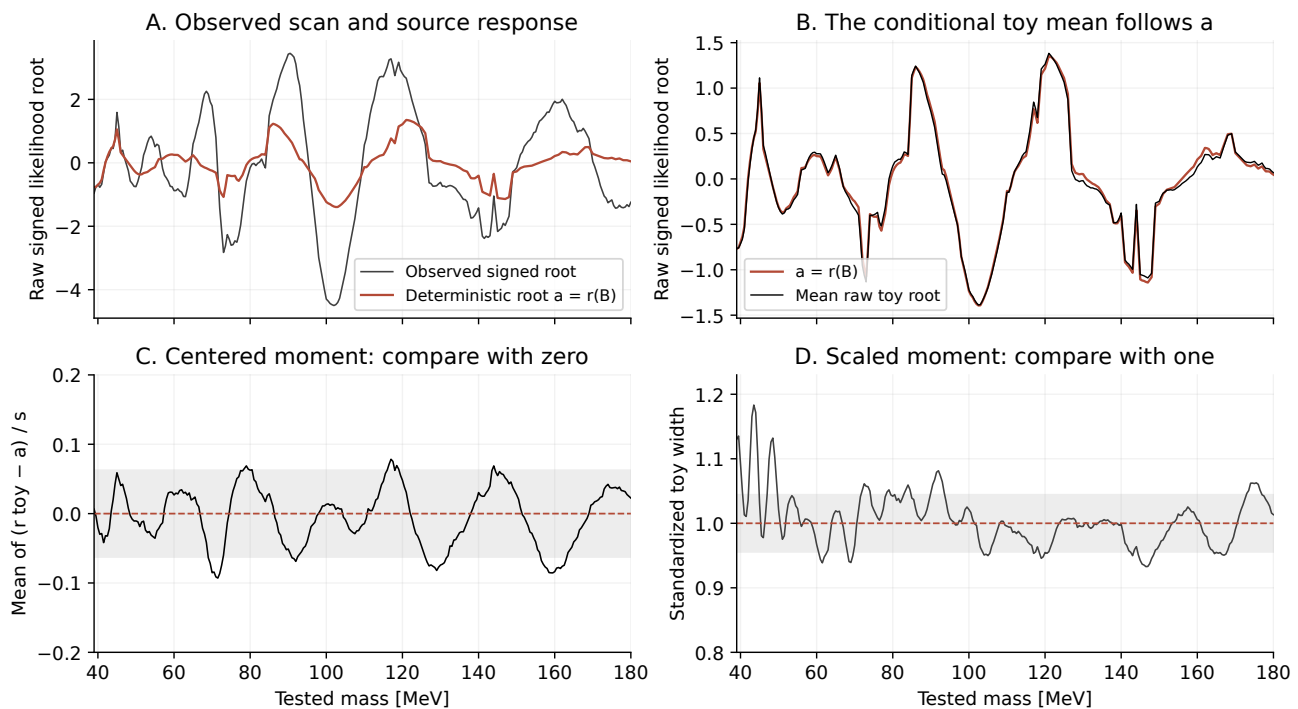


Figure 4: A clearer rendering of the 2016 part of v5.8.2 Figure 3. Panel A adds the unchanged observed scan; panel B compares the raw toy mean with the deterministic offset. Panels C and D then show the centered mean and standardized width separately. Gray bands mark only approximate pointwise one-standard-error scales for 256 independent Gaussian draws, $1/\sqrt{256}$ and $1/\sqrt{2(255)}$; they are not simultaneous acceptance bands. The mass coordinates are correlated, and all curves condition on the same fixed source.

6 Source learning and signal recovery are different tests

The signal-learning concern is nevertheless supported by a separate experiment. In v5.8.1, a signal was added to the *source data*, the unmasked GP source was rebuilt, and the change in B was measured. The nominal 2016 source absorbed 88.7% and 88.4% of the added counts inside the 76 and 90 MeV windows, respectively, and about 72% of the signal-shaped component. By contrast, adding signal to an already frozen nominal source and running the masked extraction recovered roughly 96% of the injected amplitude. Successful extraction recovery does not qualify a source that is subsequently rebuilt through the signal. The absorbed fraction is a deterministic transfer coefficient, not a detection efficiency, false-positive rate or a statement that this fraction of the actual observed excess is signal.

The practical concern is therefore the full procedure data $\rightarrow \hat{B} \rightarrow$ reference calibration, not merely the masked extraction. A learned feature can enter the putative null, affect its deterministic response and covariance, and change the resulting score. The source estimate was held fixed in the v5.8.2 toys, so their agreement tests propagation *conditional on that estimate*; it does not test repeated source learning or establish a background-only physical truth. A smaller stress RMS is useful for diagnosing compatibility between two modeling operations, but cannot choose the background model by itself.

We retain $\pm 2.25\sigma_m$ throughout. Widening only the extraction mask to $3\sigma_m$ would leave the unmasked source fit untouched, so it would not cure this source-learning mechanism. For an ideal Gaussian template, the full tails beyond these half-widths are about 2.445% and 0.270%, respectively; those template-tail fractions are distinct from the much larger source-absorption numbers. No new width scan was performed.

7 The proposed high-psum 1% source: a bounded test

A real candidate is available: the 2021 file `final_1pct_invM_v7.root`, histogram `preselection/h_invM_psumgt2p8_8000`. Its key explicitly labels $p_{\text{sum}} > 2.8$; the stored histogram contains 6,336,937 visible counts. The existing selection audit finds no event identities, run list, prompt-TC certificate or exact luminosity ratio. Its counts are 4.476% of the native-10% histogram overall, and 5.148% in 87–97 MeV. These are count ratios, not measured exposure ratios. The native-10% input comes from the documented prompt-TC production; the high-psum key alone does not establish matching reconstruction or all other cuts. This is a 2021 control and cannot supply the full-2016 offset without a separate, validated cross-dataset transfer model.

We tested the proposal as an explicitly conditional calculation. The high-psum histogram was exactly rebinned to the current 422-bin support, smoothed using the existing 2021 source kernel at 76 MeV, and multiplied by ten. The same current extraction, resolution and $\pm 2.25\sigma_m$ masks were then used at 201 integer masses from 50 to 250 MeV. The three sources are the native high-psum GP mean, that identical mean multiplied by ten, and the saved native-10% GP mean. This is a common-method comparison, not a validation of the nominal exposure transfer. No kernel was selected to improve these results.

The deterministic RMS offsets are 0.0554, 0.0283 and 0.2576 for those three sources, respectively. Offset scaling is not necessarily $\sqrt{10}$: reconditioning the extraction at the new count level changes its GP noise and interpolation. We also generated 64 complete independent Poisson spectra for each target-scale source and fitted six fixed anchors, 65, 76, 90, 92, 120 and 180 MeV, for 768 fits. The same draw is reused at all six masses. Centered means span -0.188 to 0.187 , and standardized widths span 0.914 to 1.080, consistent with only a modest moment check. These six-anchor toys are not a full-search calibration or a rare-tail measurement; their approximate mean and width Monte Carlo errors are 0.125 and 0.089. All 603 deterministic and 768 toy fits satisfy the retained 2×10^{-7} score criterion.

An additional injection test addresses the key physical issue. At 76 and 90 MeV, add a five-error target-scale Gaussian signal to the fixed $10B_{1\%}$ spectrum. The extraction recovers 96.67% and 96.68% of its amplitude. Separately, add one tenth of that same signal to the *1% source counts*, rebuild the unmasked GP, and multiply the new source by ten. The source absorbs 84.04% and 77.83% of the target window yield, or 69.02% and 63.84% of its Poisson-weighted matched shape. Thus a smaller, statistically separate sample would help limit reuse of the search fluctuations, but a genuine signal present in both samples can still be learned as background.

2021 high-psum 1% control: conditional source transfer at fixed $\pm 2.25\sigma$

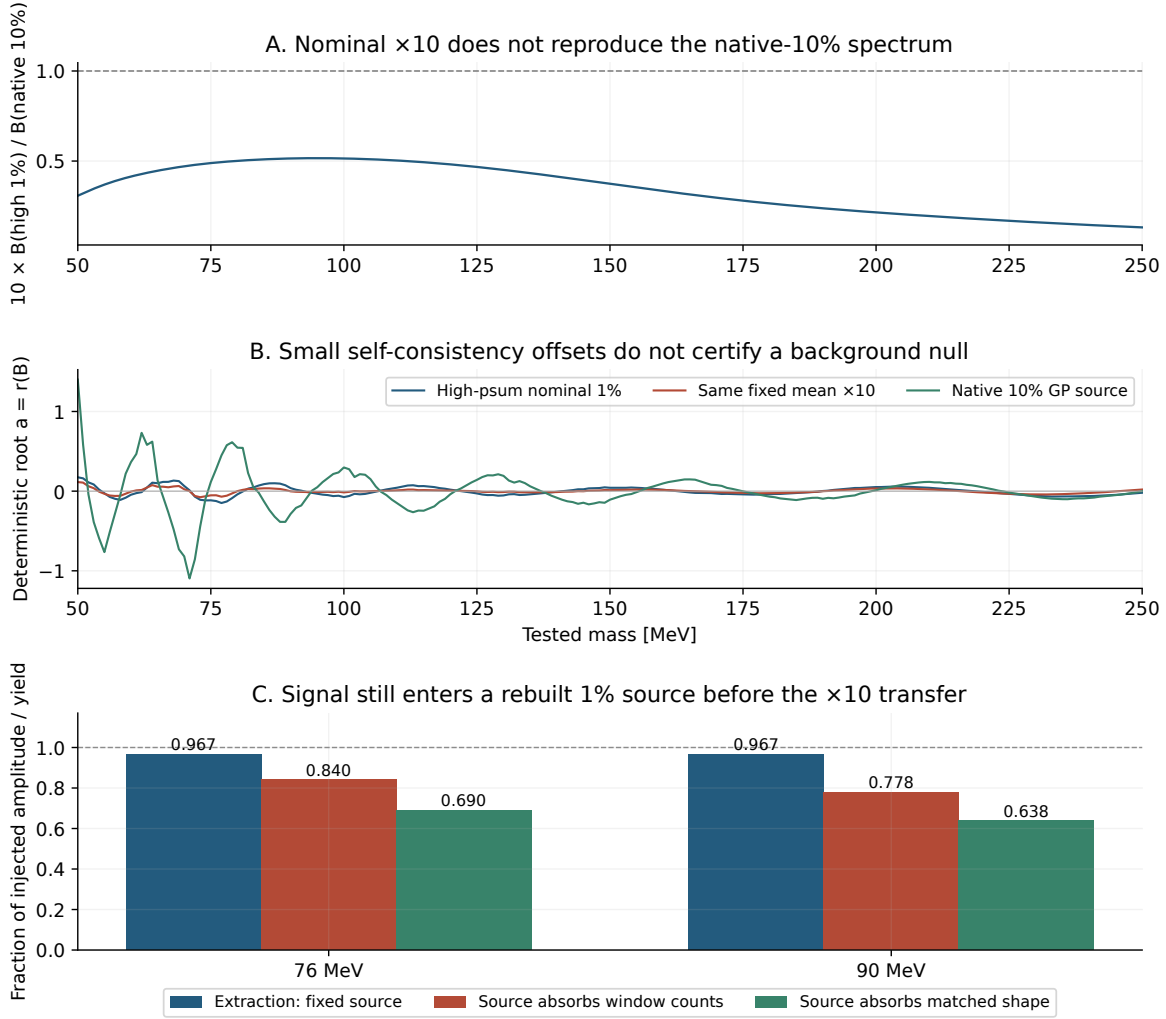


Figure 5: New fixed-width 2021 controls. A: the nominally scaled high-psum source is only about 13–52% of the native-10% GP mean over 50–250 MeV, with substantial shape dependence. B: deterministic source responses using the same extraction settings. C: fixed-source recovery versus source rebuilding after a known signal injection. The two source-learning bars use different projections of the same rebuilt mean; neither is a probability. The smaller offsets in B do not repair the transfer mismatch or the absorption in C.

8 What would make a 1%-to-10% calibration defensible?

The useful next experiment is a prediction from an audited control sample, with its uncertainty, rather than a fixed smoothed histogram treated as exact. First verify the event selection, processing version, exposure and event overlap. If the 1% is included in the 10%, either reserve a disjoint search sample or model their joint sampling; the two are not independent merely because the toy random seeds differ. Freeze the source construction, search statistic and mass grid before inspecting the validation sample. Assess continuum interpolation and source-injection absorption across the declared domain, including boundaries, and retain a signal-aware source model or controls where absorption is acceptably bounded.

Source uncertainty matters especially when increasing the exposure. If a disjoint source has mean counts λ and the target has mean $k\lambda$, the deliberately simple unsmoothed example $N_{\text{target}} - kN_{\text{source}}$ has variance $(k + k^2)\lambda$, whereas treating the source as fixed keeps only $k\lambda$. A nested source instead

contributes a covariance term. This example is not the variance of the actual GP statistic: smoothing and the extraction response must propagate the relevant modes, or an outer ensemble must repeat source estimation and target generation. For each specified underlying background truth, jointly generate control and target samples with the correct exposures and event overlap. Refit the source from that control realization, apply the declared transfer and calibration, and scan the corresponding target realization. Repeat every data-dependent choice that would be made in the real analysis. Generating the target from each noisy fitted source instead would conceal the source-estimation error that this test is intended to expose. Use separate realizations to validate local null tails, signal recovery and the maximum distribution. The existing fixed-source GP/Poisson agreement is a valuable inner check within this larger procedure.

The present high-psum calculation therefore supports keeping this source as a sensitivity control and identifies what must be qualified next. It does not justify replacing the released local significances with a new offset, certifying source independence, or adopting the source merely because its deterministic response is small.

9 Statistical fairness: two GPs, two different jobs

The background GP estimates the expected counts used by the likelihood. The significance-field GP approximates the joint null distribution of the resulting scan. They do not need to be statistically independent: the second must reproduce the response of the first, including its sideband training and profiling. Calling both objects GPs neither establishes nor invalidates the significance calculation. The substantive question is whether the generating background and the complete analysis procedure represent the null being tested.

In this application let $r_j(n)$ be the signed profile-likelihood root at mass m_j , and let B denote a fixed generating spectrum. The saved response is

$$r_j(n) \simeq a_j + \sum_b D_{bj} \xi_b, \quad a_j = r_j(B), \quad D_{bj} = \sqrt{B_b} \left. \frac{\partial r_j}{\partial n_b} \right|_B, \quad \xi_b \sim N(0, 1). \quad (1)$$

Thus $s_j^2 = \sum_b D_{bj}^2$ and $R_{ij} = \sum_b D_{bi} D_{bj} / (s_i s_j)$ describe the scale and correlation of the *fitted response*. The derivatives include the response of the GP prediction and its count-dependent covariance to fluctuations in the training bins. This is more information than the overlap of the narrow signal templates alone.

Ananiev–Read model a signed local significance field by a zero-mean, unit-variance GP, estimate its covariance from background response data, and sample its maximum to estimate the trials factor. Their comparison with an upcrossing bound concerns the same null field; it does not promise a smaller penalty than an arbitrary resolution-based Šidák count. Their finite-grid discussion also warns that coarse sampling can miss crossings and maxima [1]. The additional a_j, s_j transformation here is an HPS reference-response construction, not a new requirement imposed by that paper.

The nominal source B was fitted to the observed spectrum and then held fixed. Estimating a background from data is familiar in a parametric bootstrap and is not by itself a statistical error. The limitation is that this calculation has not calibrated the outer source-fitting and method-selection procedure, or established robustness to an absorbed signal or an inadequate background family. Independent Poisson draws validate the field approximation *conditional on that fixed source*; they do not establish independence of the source from the observed data. Favorable stress response must therefore be one diagnostic among coverage, injected-signal recovery, background adequacy and held-out behavior.

10 Keeping the raw ordering is a separate, explicit option

There is no obligation to replace the observed likelihood scan by $z_j = (r_j - a_j)/s_j$ to perform a look-elsewhere calculation. Two validly specified *conditional* questions are

$$T_r = \max_j \max(0, r_j), \quad p_{\text{global}}^{(r)} = P_B(T_r^* \geq T_r^{\text{obs}}), \quad (2)$$

$$T_z = \max_{j:r_j>0} \frac{r_j - a_j}{s_j}, \quad p_{\text{global}}^{(z)} = P_B(T_z^* \geq T_z^{\text{obs}}). \quad (3)$$

Mass-dependent centering and scaling change the ordering, so these are different tests of the same events. Neither test is to be chosen retrospectively because it yields the smaller probability. Retaining T_r preserves the original peak ranking; comparing it to its simulated null still calibrates its probability, without subtracting an offset from the observed fit.

2016, 39–180 MeV	Raw maximum T_r	Reference maximum T_z
Selected mass [MeV]	90.5	91.5
Raw root r	3.452500	3.333160
Deterministic response a	0.712849	0.496739
Response scale s	0.982030	0.982836
Reference coordinate $(r - a)/s$	2.789784	2.885955
Fixed- B Gaussian local tail	0.002637	0.001951
Gaussian global exceedances / 200,000	13,559	20,975
Gaussian global probability	0.067795	0.104875
Poisson global exceedances / 256	18	36
Poisson global probability	0.0703	0.1406
Poisson global 95% interval	[0.0422, 0.1088]	[0.1005, 0.1893]

Table 4: New comparison using the same frozen v5.8.2 fields and the same 256 complete Poisson scans. No likelihood fit or background source was changed. Intervals are Clopper–Pearson; Gaussian probabilities are conditional model approximations. Counts, k/N and add-one estimates are recorded separately in the numerical ledger.

At 90.5 MeV, $\bar{\Phi}(3.452500) = 0.0002777$ is the conventional standard-normal mapping of the raw root, whereas the fixed-source response gives $\bar{\Phi}[(3.452500 - 0.712849)/0.982030] = 0.002637$. Dividing the conditional raw-global probability by the first of these mixes a marginal-calibration change with the look-elsewhere factor. The 256 Poisson toys have zero local exceedances at this mass: they cannot resolve a local tail of order 10^{-3} , and do not prove that either precise local approximation is correct. Passing previous statistical tests supports the procedure on the nulls and observables those tests covered; it does not make a new source-conditioned centering mandatory, or guarantee every marginal tail under every new source.

11 An apples-to-apples upcrossing comparison

For the same standardized Gaussian grid, define $N_u = \sum_{j=1}^{J-1} \mathbf{1}(Z_j \leq u, Z_{j+1} > u)$. A scan exceeds u only if its first node already exceeds u , or there is an upcrossing. Hence

$$P(\max_j Z_j > u) \leq \bar{\Phi}(u) + E[N_u], \quad E[N_u] = \sum_{j=1}^{J-1} 2T\left(u, \sqrt{\frac{1 - R_{j,j+1}}{1 + R_{j,j+1}}}\right), \quad (4)$$

where T is Owen’s T function. The right-hand side is clipped at one. This identity for each adjacent crossing probability gives a bound for the *actual saved Gaussian grid*; it does not assume a differentiable

continuum kernel. All thresholds shown below exceed the positive-fit gate at every mass, so that the gate makes no change to these comparisons.

At the 2016 threshold $u = 2.885955$, the saved correlated-field tail is 0.104875 and the same-grid upcrossing bound is 0.120378. The other datasets show the same ordering: $0.02345 < 0.02527$ (2015), $0.33352 < 0.42689$ (2021), and $0.22953 < 0.27340$ (combined). This is the relevant less-conservative comparison. By contrast, the 2016 resolution-count Šidák value 0.02953 and the smooth signal-template upcrossing estimate 0.06521 describe simplified correlation models. The independent 283-node comparison gives 0.42461. The full fitted-response field lies between these illustrative answers; the smaller resolution-only answers do not establish overconservatism of that field.

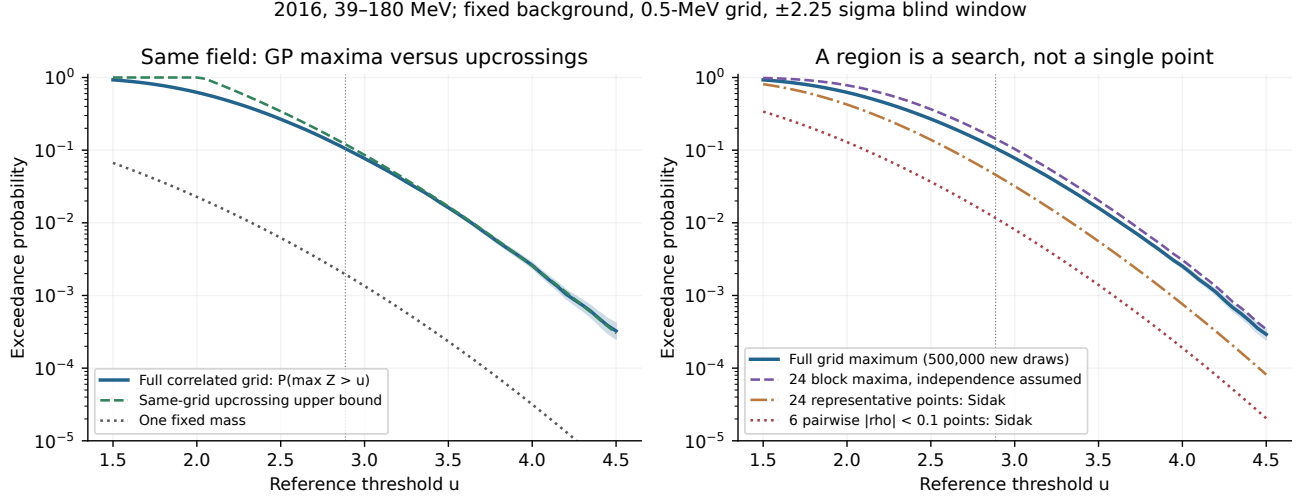


Figure 6: New 2016 comparison with the source and test held fixed. Left: the sampled Gaussian maximum is less conservative than the analytic upcrossing bound for the same finite grid. Right: counting representative points misses searches within the intervening regions; treating the properly calculated region maxima as independent errs in the other direction in this example. The left curve reuses 200,000 saved fields; the right uses 500,000 fresh fields with the same covariance. Shading denotes binomial 95% intervals for the joint maxima. These are fixed-background diagnostics.

12 What a 10% correlation criterion can and cannot do

For N independent, identically calibrated point tests, $p_{\text{global}} = 1 - (1 - p_{\text{local}})^N$. A tolerance $|\rho| < 0.1$ is not independence, and selecting weakly correlated points does not preserve the maximum of the masses omitted between them. The saved 2016 field makes both issues concrete. A left-to-right rule that selects the next mass with $|\rho| < 0.1$ relative to the previous selected mass yields 24 points. Their non-neighbor correlations still reach $|\rho| = 0.835$. Requiring the bound against *every* selected point yields six points when starting from the low-mass end and seven when starting from the high-mass end. The count depends on the selection rule. Every adjacent native-grid correlation is at least 0.889, so no split into two contiguous sets makes all correlations across the boundary smaller than 0.1.

A defensible regional version would first compute

$$p_k(u) = P\left(\max_{m \in \mathcal{R}_k} Z(m) > u\right), \quad p_{\text{independent regions}}(u) = 1 - \prod_k [1 - p_k(u)], \quad (5)$$

and then test independence of the *regional maxima*. The maximum within each region must already include its internal look-elsewhere effect. Using midpoint blocks around the 24 representative masses, the new conditional simulation gives a full-grid tail $53,201/500,000 = 0.10640$ (95% interval

[0.10555, 0.10726]). Multiplying the regional non-exceedance probabilities gives 0.14341; pretending each region is one normal point gives 0.04579. The six-point all-pairs rule gives 0.01165. None is a substitute for the already available joint-field maximum.

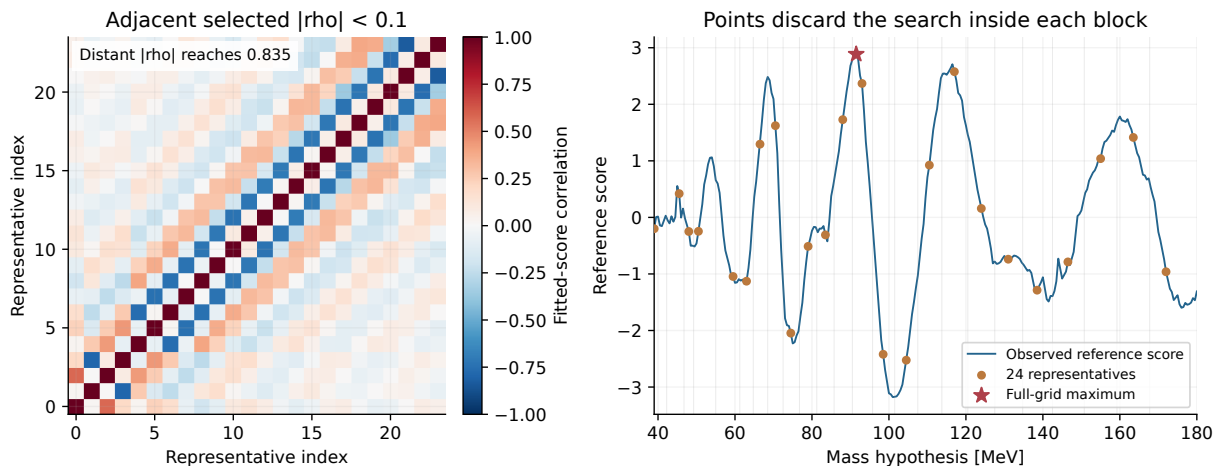


Figure 7: The proposed threshold made explicit. Left: low correlation with the immediately preceding selected point does not prevent strong correlations with more distant selected points. Right: 24 representative points do not capture the full observed scan or its maximum. Vertical lines delimit one illustrative contiguous partition, not independently established signal regions.

The tail-equivalent count $N_{\text{eff}}(u) = \log[1 - p_{\text{global}}(u)] / \log[1 - \bar{\Phi}(u)]$ is a useful way to quote the answer from the full field; it is generally threshold dependent. For the saved 2016 peak it is 56.73, whereas the spectral participation rank is 9.16 and the grid contains 283 nodes. These measure different things. For a smooth stationary field, the high-threshold crossing term scales as $\exp(-u^2/2)$ while the local tail scales approximately as $\exp(-u^2/2)/u$, already explaining why a single constant effective count need not describe all thresholds. A discovery-threshold penalty also does not, by itself, change a fixed-mass exclusion limit; a new discovery reach requires signal-injection power for the specified global test.

The initial 100,000-draw block check had an unusually high tail of 0.10910. The covariance reconstruction and positive-fit gate were verified, and the same seed was extended to 500,000 draws. The result above differs from the saved 200,000-draw estimate by about 1.9 combined Monte Carlo standard errors; their 95% intervals overlap. This sampling check is recorded explicitly rather than being interpreted as a change to the physical model or a calibrated source uncertainty.

13 What to adopt, and what still needs validation

Retain the reviewed likelihood, the raw observed scan and the fixed $\pm 2.25\sigma_m$ blind window. Retain the significance-field GP as an efficient way to simulate the null response of that procedure once a defensible source policy is specified. Do not adopt a mass-dependent reference subtraction solely because it removes the large stress offset, and do not replace the full scan by a count of nominally decorrelated points.

The deterministic offset remains a valuable diagnostic. It makes a source-extractor mismatch visible. The appropriate response is to investigate the source and its uncertainty, and to test the resulting analysis on independent simulated experiments and signal injections. Subtracting the offset defines a different test; it does not demonstrate that the original mismatch has been understood. Conversely, declining to subtract it does not establish that the raw roots have standard-normal null tails.

13.1 A concrete next calibration

A useful next study would change the source qualification while holding the extraction window fixed:

1. Establish the event selections, normalization and overlap for a control sample from the same campaign as the search. If the nominal 1% sample is contained in the 10% search, using it as an independent control requires a disjoint search sample and its correct exposure. The available 2021 high- p_{sum} histogram cannot determine a 2016 offset without a separate transfer model.
2. Specify one coherent generative experiment for each full scan. A control-to-search transfer may require a mass-dependent factor t_j , not the number ten: $B_{\text{search},j} = t_j B_{\text{control},j}$. Qualify uncertainty in the control fit and in this transfer. Estimating nuisance parameters from observed data is allowed; freezing their fitted values only answers a conditional question.
3. Validate the complete proposed procedure under several plausible background truths. When source construction or calibration uses the search data, repeat those steps in each outer pseudoexperiment and evaluate the resulting reported p-values. Use separate validation draws from those used to construct the response or calibrate tails. If the control is independent, simulate the control and search jointly when evaluating unconditional behavior.
4. Inject signals before source estimation as well as after it. The first test measures what the source-building policy can learn; the second measures the extractor with a frozen source. Check size, signal recovery and uncertainty across mass before selecting a source or ordering. Then test mass-grid convergence and tails at the precision required by the scientific claim.

This specifies a testable program rather than a reason to choose whichever reference produces the highest observed significance. Choosing among source models, combination methods or mass domains after inspecting their p-values creates an additional selection question beyond a mass-only trials correction.

13.2 Discovery reach and exclusion reach are different quantities

A lower global discovery significance does not automatically weaken the existing fixed-mass 90% CL_s exclusion limit. Discovery and exclusion use different tail events[3]. Pointwise exclusion curves are not obtained by applying a discovery trials factor to each endpoint. A simultaneous confidence statement across mass would require its own construction.

For discovery reach, fix a false-positive level α , calibrate the threshold c_α of the declared full-search statistic T , and evaluate

$$P_{B+S(\mu, m_0)}(T \geq c_\alpha), \quad P_B(T \geq c_\alpha) \leq \alpha.$$

The signal-plus-background experiments must follow the same source-building policy. No global p-value ratio alone determines a coupling reach. The present report therefore establishes neither a new exclusion curve nor an improved discovery reach.

13.3 Combined searches: the useful result is model dependence

The common-coupling fit, Fisher combination, signed Stouffer combination and independent-amplitude common-mass likelihood ask different questions. Their different preferred masses are understandable without attributing extra independent evidence to a single spectrum. The v5.8.5 free-amplitude result at 92 MeV remains a conditional moderate excess; the same release found that the inspected clustering of peaks was not unusually coherent under its chosen null. The comparison is useful for understanding the data, but does not authorize selecting the largest of those significances as the discovery result.

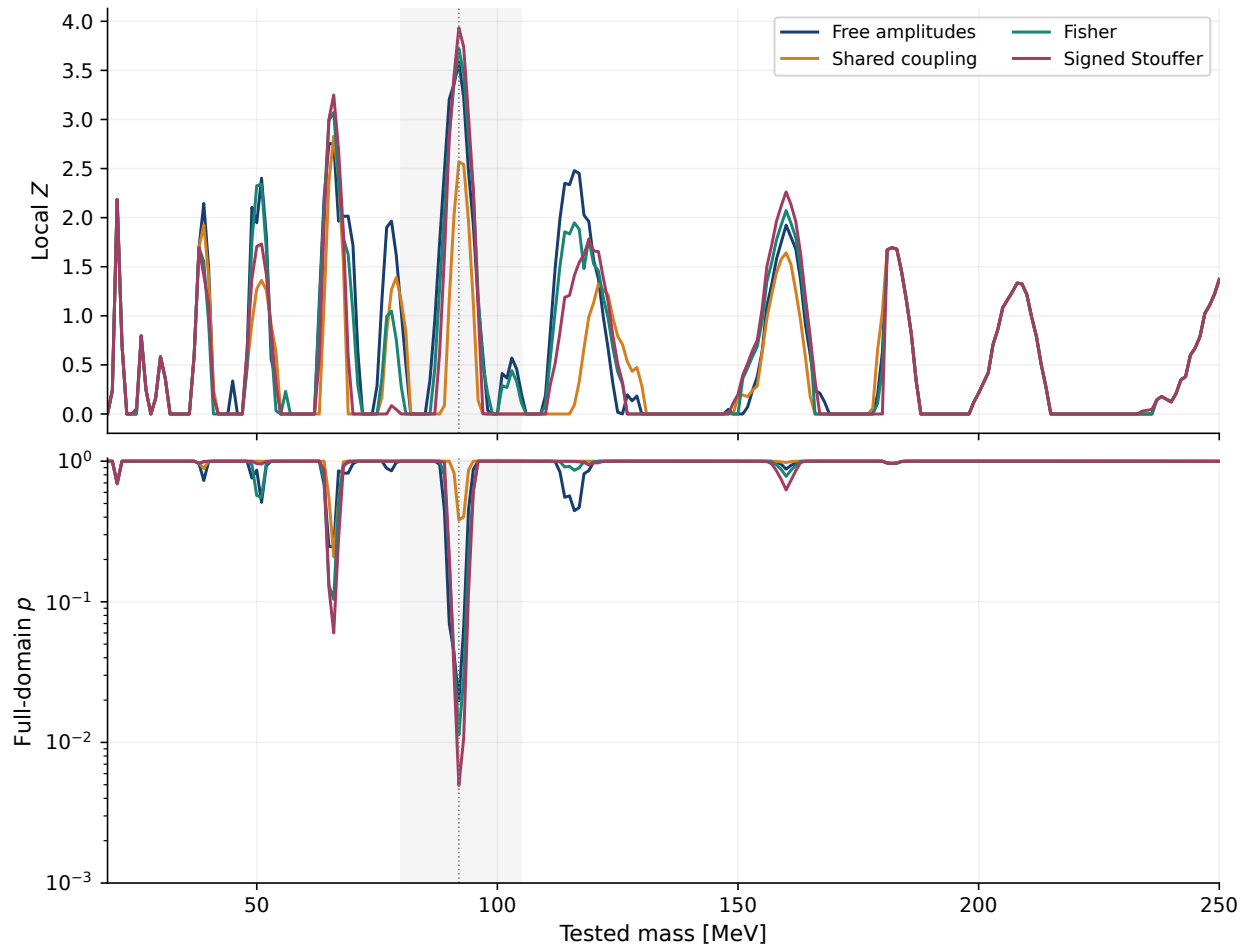


Figure 8: Baseline comparison from the earlier v5.8.5 report. All curves use the $\pm 2.25\sigma_m$ mask and the 1 MeV mass grid; the global probabilities refer to the full 19–250 MeV domain. The shaded interval is a post-observation diagnostic zoom. The shared-coupling and independent-amplitude likelihoods impose different physical restrictions, while Fisher combines gated local probabilities and signed Stouffer combines continuous signed reference scores. These curves are strongly related uses of the same events.

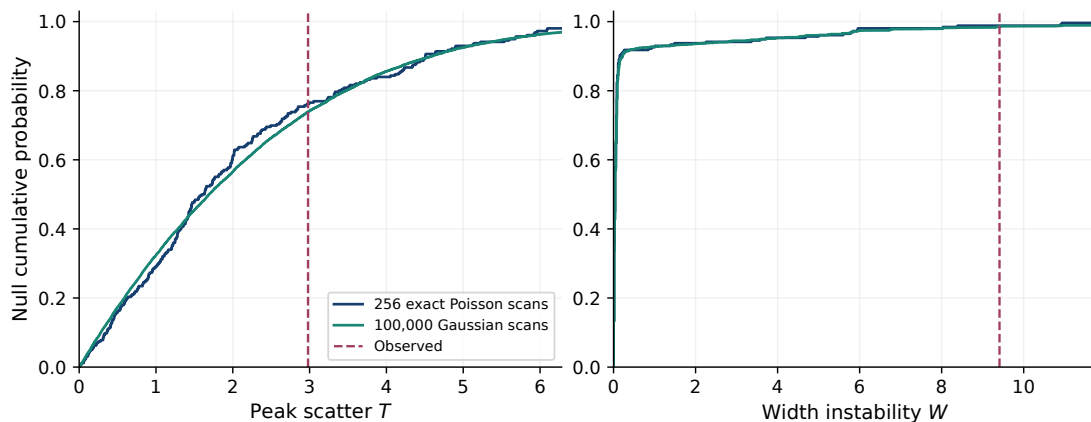


Figure 9: The earlier v5.8.5 coherence check on 80–100 MeV. The lower tail, not the upper tail, measures unusually close peak locations. The observed clustering gives 195/256 direct lower-tail counts, or add-one probability 0.76265, and is not unusual under the fixed-source null. The four historical widths enter this archived sensitivity diagnostic; this review makes no new width choice.

14 Scope, provenance and reproducibility

The numerical claims are separated in the package into archived results and new diagnostics. The new work consists of the retained-raw-ordering audit from saved ensembles; the ten-percent-correlation and regional-maximum study; and a bounded 2021 source-transfer and injection study. The archived v5.8.0–v5.8.5 sources, fields, ledgers and important plots were read directly. The referenced v5.0.5 sections and arXiv paper were checked against their actual text.

The response-field comparison uses the saved 200,000 Gaussian maxima and 256 complete Poisson scans per scope. The regional diagnostic adds 500,000 correlated Gaussian fields for 2016 and compares the same covariance, threshold and domain. Its Monte Carlo interval is distinct from that of the saved ensemble. The new 2021 study uses 603 deterministic mass/source states, 64 coherent six-anchor Poisson experiments for each of two sources, and two source-rebuilding injection anchors. These are local diagnostics, not a new full-domain Poisson calibration.

The source bundle contains the LaTeX report, vector figures, the new scripts and ledgers, the precise high- p_{sum} input histogram, copied engine dependencies, and the frozen fields used in the statistical audit. Commands in `README.md` rebuild the report and the new calculations separately. The frozen files needed to render the report are included; historical studies are represented by their saved outputs and code snapshots, rather than a promise that every historical fit can be rerun from this smaller bundle.

A SHA-256 parent manifest records 3,782 original files. The closing audit checks that these released files are unchanged. Report verification includes numerical assertions, PDF text extraction, rendered-page inspection and an isolated rebuild. The work was constrained to one hour and at most 20 additional percentage points of account-wide weekly usage; the measured completion time and usage snapshots are recorded in the delivery metadata.

References

- [1] V. Ananiev and A. L. Read, *Gaussian Process-based calculation of look-elsewhere trials factor*, JINST 18 (2023) P05041, arXiv:2206.12328v3, especially Sections 2, 2.1 and 2.2.
- [2] E. Gross and O. Vitells, *Trial factors for the look elsewhere effect in high energy physics*, Eur. Phys. J. C 70 (2010) 525–530, arXiv:1005.1891.
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- [4] HPS-GPR, *Analysis Note v5.0.5*, 16 September 2026, Sections 4.20 and 6.11, pinned local source and PDF.
- [5] HPS-GPR, studies v5.8.0–v5.8.5, 17–19 September 2026. Exact input identities, scripts, numerical ledgers and selected figures are included in the accompanying source-and-data package.